

ALGEBRAIC COMBINATORICS LEARNING SEMINAR FALL 2026
TALK 1: INTRODUCTION TO SOLVABLE LATTICE MODELS (SEP 16)

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1. INTEGRABILITY

1.1. Classical mechanics. The word “integrable” comes from the problem of integrating differential equations.

In classical mechanics, the central question was: when can we explicitly solve the equations of motion?

Let M be the phase space of dimension $2n$. Let $A : M \rightarrow \mathbb{R}$ be a function, also known as observable, or physical quantity. The Hamiltonian equation gives the equation of motion

$$\{H, A\} = \dot{A}$$

where H is the Hamiltonian function, i.e. the energy.

The most important philosophy found in the last century (due to Neother) is
conserved quantity (some function A with $\dot{A} = 0$) $\iff$ symmetry (some group G acts on M nicely).

For example,

Energy $\iff$ Time translations
Momentum $\iff$ Translations
Angular momentum $\iff$ Rotations

In mathematical language, it says if a group G acts (nicely) on M , then it has moment map $M \rightarrow \mathfrak{g}^*$. Then any linear map $\mathfrak{g}^* \rightarrow \mathbb{R}$ provides a conserved quantity. Conversely, if $\dot{A} = 0$, then it locally arises from a momentum map.

If a system has a conserved quantity, the complexity can be reduced by one. Since A is conserved, it suffices to restrict to $A^{-1}(a) \subset M$ for some a . Since a group G acts on $A^{-1}(a)$, we can further take quotient $A^{-1}(a)/G$. Mathematically, this is the procedure of the Hamiltonian reduction.

If a system has two conserved quantities A, B , the complexity can be reduced by two, assuming $\{A, B\} = 0$. The condition is to make sure B is well-defined over the Hamiltonian reduction.

Integrable system is the case when the complexity can be reduced as few as possible.

Definition. An **integrable system** on a symplectic manifold M of dimension $2n$ is a collection of smooth functions $H_1, \dots, H_n$ on M such that they are in involution (i.e., $\{H_i, H_j\} = 0$), and the differentials dH_i are linearly independent on a dense open set in M .

Nowadays, integrability is measured by the number of functions in involution.

Integrable $\iff$ how many functions are in involution.

1.2. Quantum Mechanics. In quantum mechanics, everything is the same, but we should replace all Poisson brackets by Lie brackets. Now, quantum integrability is measured by how many elements commute.

quantum integrable $\iff$ how many elements are commutative

Example. Consider the operator u_i (for $i \geq 0$) on

$$\mathcal{F} = \bigoplus_{\lambda \text{ is a partition}} \mathbb{Q} \cdot \lambda.$$

by

$$u_i \lambda = \begin{cases} \lambda \cup \text{a box in the } i\text{-th row,} \\ 0, & \text{if impossible.} \end{cases}$$

Consider

$$e_r = \sum_{i_1 < \dots < i_r} u_{i_r} \cdots u_{i_1}.$$

Then

$$[e_r, e_s] = 0.$$

Example. Inside the affine Hecke algebra $\hat{H}_W$, we have a large commutative algebra. For any coweight λ , we define

$$Y^\lambda = T_{t_{\lambda_1}} T_{t_{\lambda_2}}^{-1}$$

by writing $\lambda = \lambda_1 - \lambda_2$ for two strictly dominant λ_i . We have

$$[Y^\lambda, Y^\mu] = 0.$$

Actually, they generate a group ring $\mathbb{Q}_t[\mathbb{Q}^\vee]$.

Note that $\hat{H}_W$ acts on Laurent polynomial ring $\mathbb{Q}_t[P]$ via Demazure–Lusztig operators. The operators corresponding to Y^λ are known as Macdonald operators. By degeneration, we can obtain the Dunkl operators.

2. SOLVABILITY

2.1. Statistical mechanics. Statistical mechanics describes a system by assigning probabilities to its microscopic configurations.

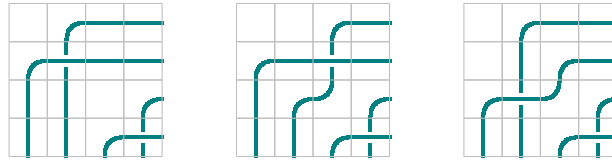
$$\mathbb{P}(\text{State} = \sigma) = \frac{1}{Z} e^{-E(\sigma)/kT}, \quad Z = \sum_{\sigma \in \Omega} e^{-E(\sigma)/kT}.$$

This is a result of the principle of maximum entropy.

When energy is a sum of local contributions, the weight becomes a product. This is precisely the structure we use in combinatorics.

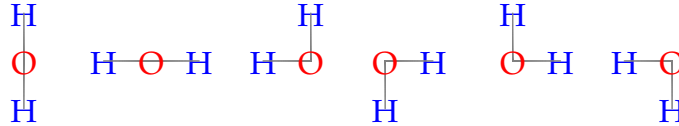
$$Z = \sum_{\sigma \in \Omega} \prod_v \text{wt}_v(\sigma)$$

Example — BPDs. The bumpless pipe dream model for (double) Schubert polynomials.

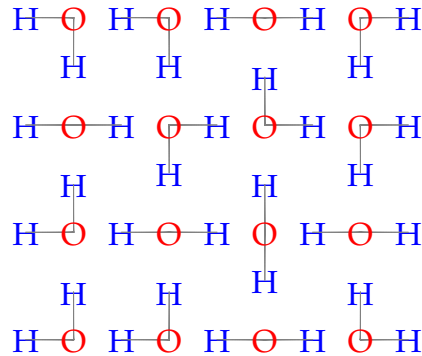


$$\mathfrak{S}_{2143}(x; y) = (x_1 - y_1)(x_3 - y_3) + (x_1 - y_1)(x_1 - y_2) + (x_1 - y_1)(x_2 - y_1).$$

Example — Planar ice. The classic example of 6-vertex model is **planar ice**.

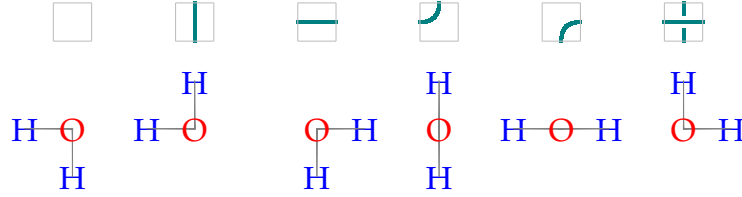


Adjacent O and H are joined either by a hydrogen bond or a chemical bond.

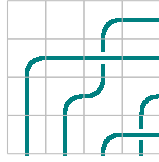


This is the first example of 6-vertex model.

Exercise. There is a bijection between planar ice and BPD:



For example, the above planar ice gives



Baxter introduced the Yang–Baxter equation and applied it to Ice-type models. The main trick is the Yang–Baxter equation and transfer matrices.

Actually Baxter characterizes when the model can be solved exactly (instead of asymptotically). So this is called “exactly solvable model” or “solvable lattice model”.

2.2. ASM enumeration. An alternating sign matrix is a matrix with entries in $\{0, 1, -1\}$ such that the nonzero entries in each row and column alternate in sign and sum to 1. For example,

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

The general enumeration is

$$\#\{\text{ASM's of size } n \times n\} = \prod_{j=0}^{n-1} \frac{(3j+1)!}{(n+j)!} = \frac{1! \cdot 4! \cdot 7! \cdots (3n-2)!}{n!(n+1)! \cdots (2n-1)!}.$$

For example, when $n = 3$,

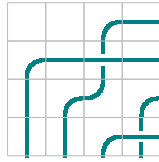
$$\frac{1!4!7!}{3!4!5!} = 7, \quad 6 \text{ permutations matrix and one } \begin{bmatrix} 0 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Zeilberger gave the first proof. Kuperberg subsequently found a much shorter proof using six-vertex integrability.

The first step of Kuperberg’s proof is the following:

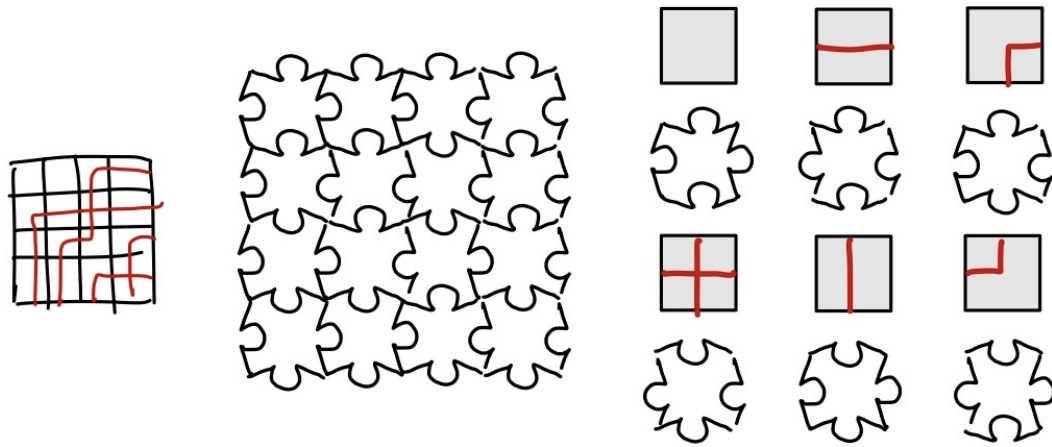
- replace 1 by , replace -1 by ;
- replace the remaining 0’s by , ,  and .

For example, the above example gives



The problem reduces to a problem of enumerating certain BPDs.

Exercise. Check this is also a 6-vertex model:



3. THE MAIN TRICK

The main trick is the Yang–Baxter equation and transfer matrices. Let us consider

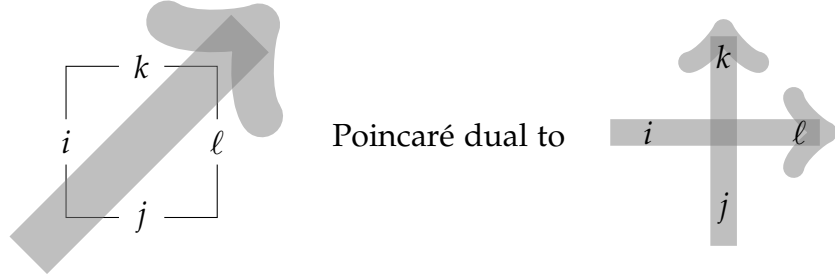
$$\mathbb{V} = \mathbb{C}e_0 \oplus \mathbb{C}e_1, \quad \begin{array}{l} e_1 = \text{a pipe,} \\ e_0 = \text{no pipe.} \end{array}$$

Consider a linear transformation (called R -matrix)

$$R : \mathbb{V}^{\otimes 2} \rightarrow \mathbb{V}^{\otimes 2}$$

$$\begin{aligned} e_1 \otimes e_1 &\mapsto \begin{array}{|c|} \hline \text{+} \\ \hline \end{array} e_1 \otimes e_1 \\ e_1 \otimes e_0 &\mapsto \begin{array}{|c|} \hline \text{┐} \\ \hline \end{array} e_1 \otimes e_0 + \begin{array}{|c|} \hline \text{—} \\ \hline \end{array} e_0 \otimes e_1 \\ e_0 \otimes e_1 &\mapsto \begin{array}{|c|} \hline \text{┌} \\ \hline \end{array} e_0 \otimes e_1 + \begin{array}{|c|} \hline \text{—} \\ \hline \end{array} e_1 \otimes e_0 \\ e_0 \otimes e_0 &\mapsto \begin{array}{|c|} \hline \text{ } \\ \hline \end{array} e_0 \otimes e_1 \end{aligned}$$

The diagram is



Then,

$$\text{assumed to be empty} \rightarrow \boxed{?} \boxed{?} \boxed{?} \dots \dots \dots \boxed{?} \boxed{?} \boxed{?} \leftarrow \text{assumed to be empty}$$

can be viewed as a linear transformation.

$$T : \mathbb{V}^{\otimes n} \longrightarrow \mathbb{V}^{\otimes n}.$$

Obtained by

$$T = p_n R_{n-1,n} R_{n-2,n-1} \cdots R_{1,2} i_1$$

where $i : \mathbb{C}e_1 \subset \mathbb{V}$ the inclusion and $p : \mathbb{V} \rightarrow \mathbb{C}e_1$ the projection.

Different rows might have different weights; we write $T = T(u)$ to denote the dependence. To solve the planar ice or enumerate the ASMs, we need the commutativity (quantum integrability)

$$[T(u), T(v)] = 0.$$

Since this is the condition for solving the model exactly, this is also called the *exactly solvable model*, or *solvable lattice models*.

The main trick found by Baxter is,

$$\text{Yang-Baxter equation} \Rightarrow \text{commutativity of transformation matrix}$$

Let me roughly explain the idea:

$$\begin{aligned}
 & \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \begin{array}{c} | \\ | \\ \dots \\ | \\ | \end{array} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} = \frac{1}{\text{wt}(\overset{\circ}{\chi}_{\overset{\circ}{\chi}})} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \begin{array}{c} | \\ | \\ \dots \\ | \\ | \end{array} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \\
 & \qquad \qquad \qquad = \frac{1}{\text{wt}(\overset{\circ}{\chi}_{\overset{\circ}{\chi}})} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \begin{array}{c} | \\ | \\ \dots \\ | \\ | \end{array} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \\
 & \qquad \qquad \qquad = \frac{1}{\text{wt}(\overset{\circ}{\chi}_{\overset{\circ}{\chi}})} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \begin{array}{c} | \\ | \\ \dots \\ | \\ | \end{array} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \\
 & \qquad \qquad \qquad = \dots \\
 & \qquad \qquad \qquad = \frac{1}{\text{wt}(\overset{\circ}{\chi}_{\overset{\circ}{\chi}})} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \begin{array}{c} | \\ | \\ \dots \\ | \\ | \end{array} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \\
 & \qquad \qquad \qquad = \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array} \begin{array}{c} | \\ | \\ \dots \\ | \\ | \end{array} \begin{array}{c} 0 \\ \text{---} \\ 0 \end{array}
 \end{aligned}$$